

Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Even)

Course Code & Title: 20EC279 – Mini Project I

**End Semester Autonomous Examination-
May 2024**

No. of Crédits : 2

Year / Section / Semester	II/C/IV	Batch	2022 - 2026
Project Presentation	End Semester Examination – May 2024	Date	06.05.2024
Project Batch No.	17		
Team Members	1. RAHUL P (71812202201) 2. SUDHARSHAN V (71812202238) 3. VAGULABARAN K M (71812202249)		
Project Title	VIRTUAL REALITY BASED FIRE FIGHTING AND DAMAGE CONTROL TRAINING MODULE		
Domain of the Mini Project	VIRTUAL REALITY		
Name of the Supervisor with Designation	Mrs.S.Saranya Assistant Professor(Sr.G)/ECE		
Mini Project Coordinator	Dr.R.Karthikamani, Assistant Professor(Sr.G)		

- Problem Description
- Abstract
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Problem Description :

- Firefighters often encounter situations where their ability to quickly and accurately assess and navigate complex, smoke-filled environments is hindered, leading to increased risks and reduced efficiency.
- The challenge lies in finding innovative, cost-effective, and VR training solutions that can adequately prepare firefighters for these challenging scenarios, ultimately enhancing their safety and effectiveness with relying on virtual reality technology.

Abstract:

Within a secure and regulated virtual setting, the Virtual Reality (VR) based firefighting training module provides a revolutionary approach to firefighter education. This cutting-edge program simulates fire dynamics, smoke propagation, and heat impacts to immerse learners in realistic fire scenarios. Using virtual firefighting apparatus, trainees learn critical skills such as incident command and fire suppression through interactive exercises. This Virtual Reality (VR) module eliminates the hazards and expenses associated with traditional live- fire training while improving situational awareness, operational preparedness, and skill development. It does this by offering configurable scenarios and real-time performance evaluation. All things considered, this technologically advanced method is an economical and significant way to raise the level of readiness and safety among firefighters.

Article Title, Authors, Year	Name of the Journal	Observation	Remarks
TITLE : Integrated Augmented and Virtual Reality Technologies for Realistic Fire Drill Training. AUTHORS : Hosan Kang, Jinseong Yang, Beom-Seok Ko, Bo-Seong Kim, Oh-Young Song, and Soo-Mi Choi , Sejong University, Seoul. YEAR:2023	IEEE Computer Graphics and Applications.	❖ Introduce a new fire drill training system. ❖ Combine AR and VR in a single head-mounted display. ❖ Overcome limitations in space, time, risk, and costs. ❖ Enhance effectiveness with realistic virtual fire situations. ❖ Emphasize interaction with physical training props. ❖ Create realistic fire effects in AR. ❖ Estimate object types and physical properties. ❖ Conducted studies show potential for improving training and education.	✓ Innovative Solution ✓ Realism ✓ Considerations

Article Title, Authors, Year	Name of the Journal	Observation	Remarks
TITLE : Virtual reality for immersive multi-user firefighter-training scenarios. AUTHORS : Philipp BRAUN*, Michaela GRAFELMANN, Felix GILL, Hauke STOLZ, Johannes HINCKELDEYN, Ann-Kathrin LANGE YEAR:2022	Virtual Reality & Intelligent Hardware	❖ Immersion is crucial for effective virtual training. ❖ A prototype VR scenario involves putting out a virtual fire on a container ship. ❖ It's a full-body, multi-user experience using motion-capture and VR headsets. ❖ Motion-capture VR works well for firefighting training in tough environments. ❖ The study introduces prototype for multi-user VR training for maritime firefighters. ❖ Future research should assess initial results, create more scenarios, and measure training progress.	✓ Focus of Study ✓ Technology Utilized ✓ Challenges

Article Title, Authors, Year	Name of the Journal	Observation	Remarks
TITLE : Fire Detection and Recognition Optimization Based on Virtual Reality Video Image. AUTHORS: XINCHU HUANG1 AND LIN DU 2 YEAR: 2020	SPECIAL SECTION ON GIGAPIXEL PANORAMIC VIDEO WITH VIRTUAL REALITY.	❖ Improve fire detection using video images and overcome issues with traditional methods. ❖ Use machine learning, specifically Support Vector Machine (SVM) and Rough Set (RS) theories. ❖ Combine SVM and RS for better fire detection. ❖ Enhance noise suppression and improve accurate fire detection and warnings. ❖ RS-SVM outperforms using SVM alone. ❖ Uses fire feature selection in virtual reality (VR) video images for better fire detection and recognition	✓ Optimized Parameters ✓ Methodology ✓ Benefits

Article Title, Authors, Year	Name of the Journal	Observation	Remarks
TITLE: Delivering Critical Stimuli for Decision Making in VR Training: Evaluation Study of a Firefighter Training Scenario. AUTHORS: Pedro Monteiro , Miguel Melo, António Valente, José Vasconcelos-Raposo, and Maximino Bessa YEAR:2021	IEEE Transactions on Human-Machine Systems, VOL. 51, NO. 2.	❖ Enable effective real-world performance through immersive VR training. ❖ Study assesses impact of feeling temperature in VR training scenarios. ❖ Involves a virtual scenario where trainees sense door temperature. ❖ Three scenarios: visual-only, thermal-only, and a combination of both. ❖ Measures time, correct procedures, and responses to assess effectiveness. ❖ Also evaluates presence, cybersickness, and usability.	✓ Focus on Haptic Thermal Stimulus ✓ Experimental Setup ✓ Goal of VR Training

Article Title, Authors, Year	Name of the Journal	Observation	Remarks
TITLE: Virtual reality for fire safety training. AUTHORS: Sarah Morélot , Alain Garrigou , Julie Dedieu , Bernard N'Kaoua YEAR: 2021	Computers & Education	❖ Fires can have severe economic and human consequences. ❖ Investigate the effects of immersion and the sense of presence on conceptual and procedural learning in fire safety training. ❖ Control variables such as gender, immersion tendency, computer and video game experience. ❖ Immersion promotes procedural but not conceptual learning.	✓ Limited Emotional Engagement ✓ Technical Issues ✓ Skill Transfer ✓ Maintenance and Updates

Article Title, Authors, Year	Name of the Journal	Observation	Remarks
TITLE: A virtual reality based fire training simulator integrated with fire dynamics data. AUTHORS: Moohyun Cha a , Soonhung Han b,n , Jaikyung Lee a , Byungil Choi c YEAR:2012	Fire Safety Journal	❖ Utilizes Computational Fluid Dynamics (CFD) for precise prediction of fluid phenomena. ❖ CFD data not easily applicable to real-time VR systems. ❖ Proposes data conversion techniques and a real-time framework for a fire training simulator. ❖ Simulator calculates various physical quantities (toxic gases, heat) and visible factors (smoke, flame) for realism. ❖ Creates a simulator for firefighting activities in road tunnels.	✓ Framework Components ✓ Safety Level-Based Visualization ✓ Application Testing

Article Title, Authors, Year	Name of the Journal	Observation	Remarks
TITLE: Case Studies in Fire Safety AUTHORS: Majed Almejmaja, Jeanine LM Skorinkoa , Brian J. Meachama YEAR:2017	Case Studies in Fire Safety	❖ Investigates the influence of cultural differences on building evacuation in the U.S. and Saudi Arabia. ❖ Focuses on recognition and reaction times during fire drills. ❖ U.S. populations generally respond faster to fire alarms compared to Saudi populations. ❖ Raises questions about the relevance and applicability of data collected in Western contexts to non-Western cultures in the analysis of building evacuation processes.	✓ Cognitive State during Fire Drills ✓ Cultural Variability in Evacuation Behavior

Article Title, Authors, Year	Name of the Journal	Observation	Remarks
TITLE: An Educational Platform to Improve Students' Learning Through Virtual Reality. AUTHORS: MARIAPAOLA PUGGIONI, EMANUELE FRONTONI , MARINA PAOLANTI , ROBERTO PIERDICCA YEAR: 2021	Institute of Electrical and Electronics Engineers	❖ ScoolAR is an educational AR and VR platform. ❖ Overcomes time constraints in education. ❖ Requires additional research, especially with larger classes. ❖ Demonstrated its significance during lockdowns in bridging theory and practical experience. ❖ Relevant for various educational disciplines in lifelong learning.	✓ Focus Areas ✓ Relevance ✓ Benefits

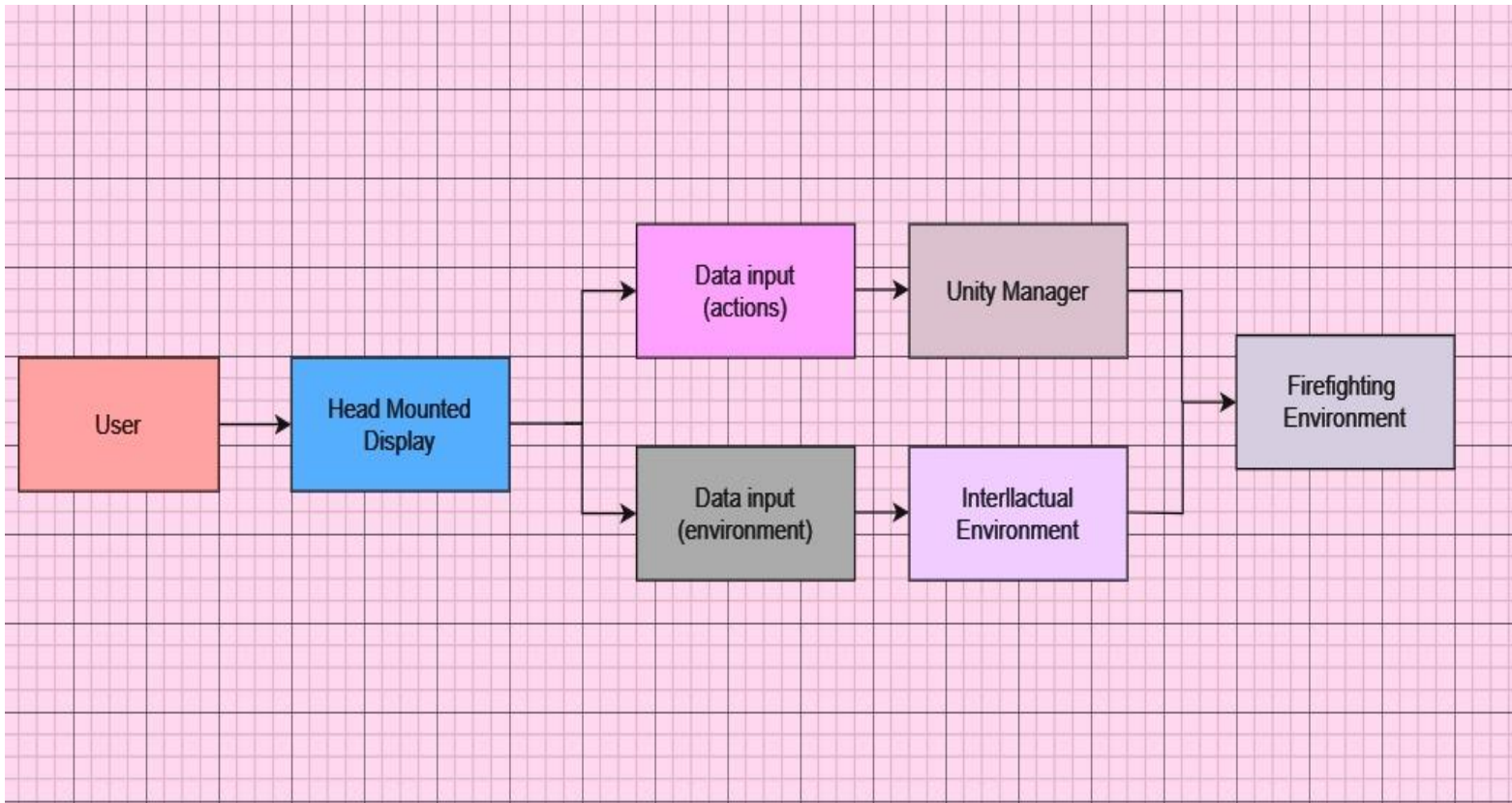
Article Title, Authors, Year	Name of the Journal	Observation	Remarks
TITLE: A Virtual Reality Dance Training System Using Motion Capture Technology. AUTHORS: Jacky C.P. Chan, Howard Leung, Jeff K.T. Tang, and Taku Komura YEAR: 2011	IEEE Transactions on Learning Technologies (Volume: 4)	❖ Presents a dance training system using motion capture and virtual technology. ❖ Captures real-time motion for remote instruction. ❖ Provides feedback to users. ❖ Future work includes automated lesson generation. ❖ Aims to improve immersion with realistic backgrounds	✓ System Description ✓ Functionality ✓ Enhancement Goals

Article Title, Authors, Year	Name of the Journal	Observation	Remarks
TITLE: Subjective QoE of 360-Degree Virtual Reality Videos and Machine Learning Predictions. AUTHORS: MUHAMMAD SHAHID ANWAR, JING WANG, WAHAB KHAN, ASAD ULLAH, SADIQUE AHMAD, ZESONG FEI. YEAR:2020	Institute of Electrical and Electronics Engineers	❖ Use machine learning to predict Quality of Experience (QoE) for 360-degree VR videos. ❖ Gathered data from user experiments. ❖ Logistic Regression model achieved 86% accuracy for perceptual quality. ❖ Introduced a Neural Network model for cybersickness. ❖ Outperformed existing methods in predicting cybersickness.	✓ Absence of Real-Time Prediction ✓ Limited Scope of QoE Aspects ✓ Data Collection

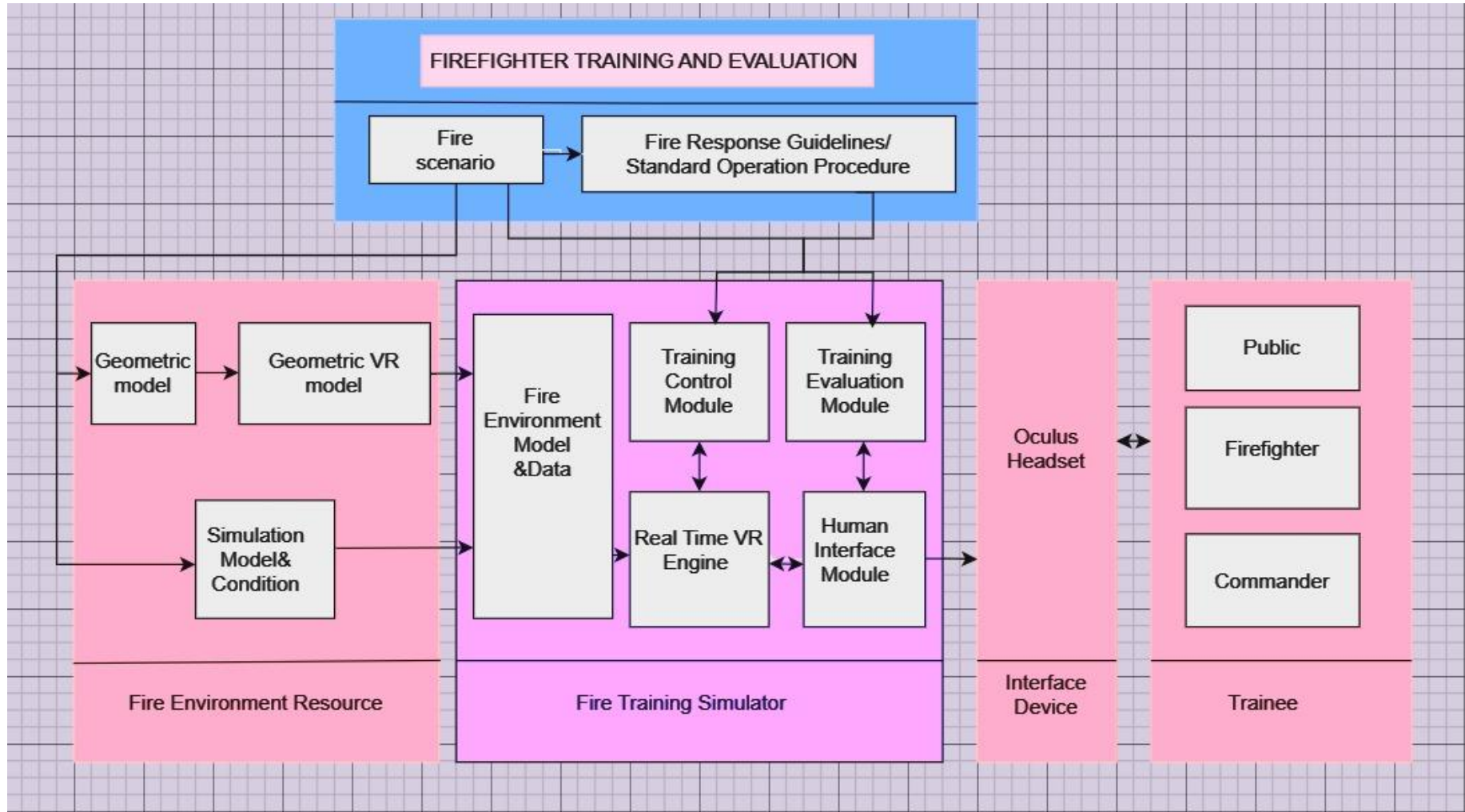
Inference :

It appears that the literary survey discussed in the provided text revolves around the use of virtual training environments for fire safety and related educational purposes. The study investigates the impact of immersion and the sense of presence on different types of learning, both procedural and conceptual, while considering various user characteristics like gender, computer experience, and video game familiarity. The findings suggest that immersion enhances procedural learning, particularly when participants have a prior understanding of fire safety. However, conceptual learning doesn't seem to benefit significantly from immersion or a sense of presence. The study also emphasizes the importance of understanding how technical devices and user characteristics affect the effectiveness of educational design, offering insight into the potential and limitations of virtual training in the context of fire safety.

Proposed Method –Block Diagram

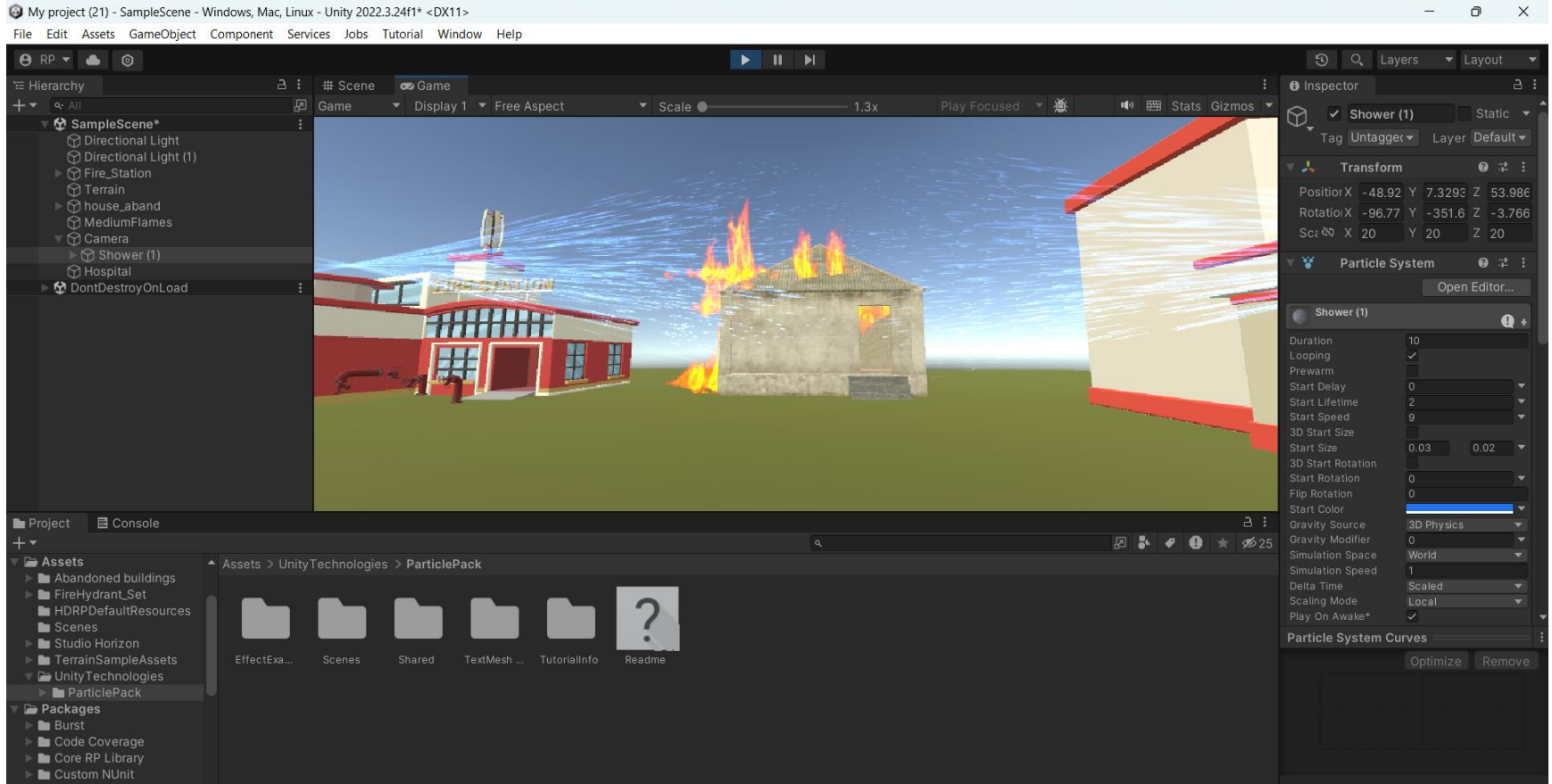


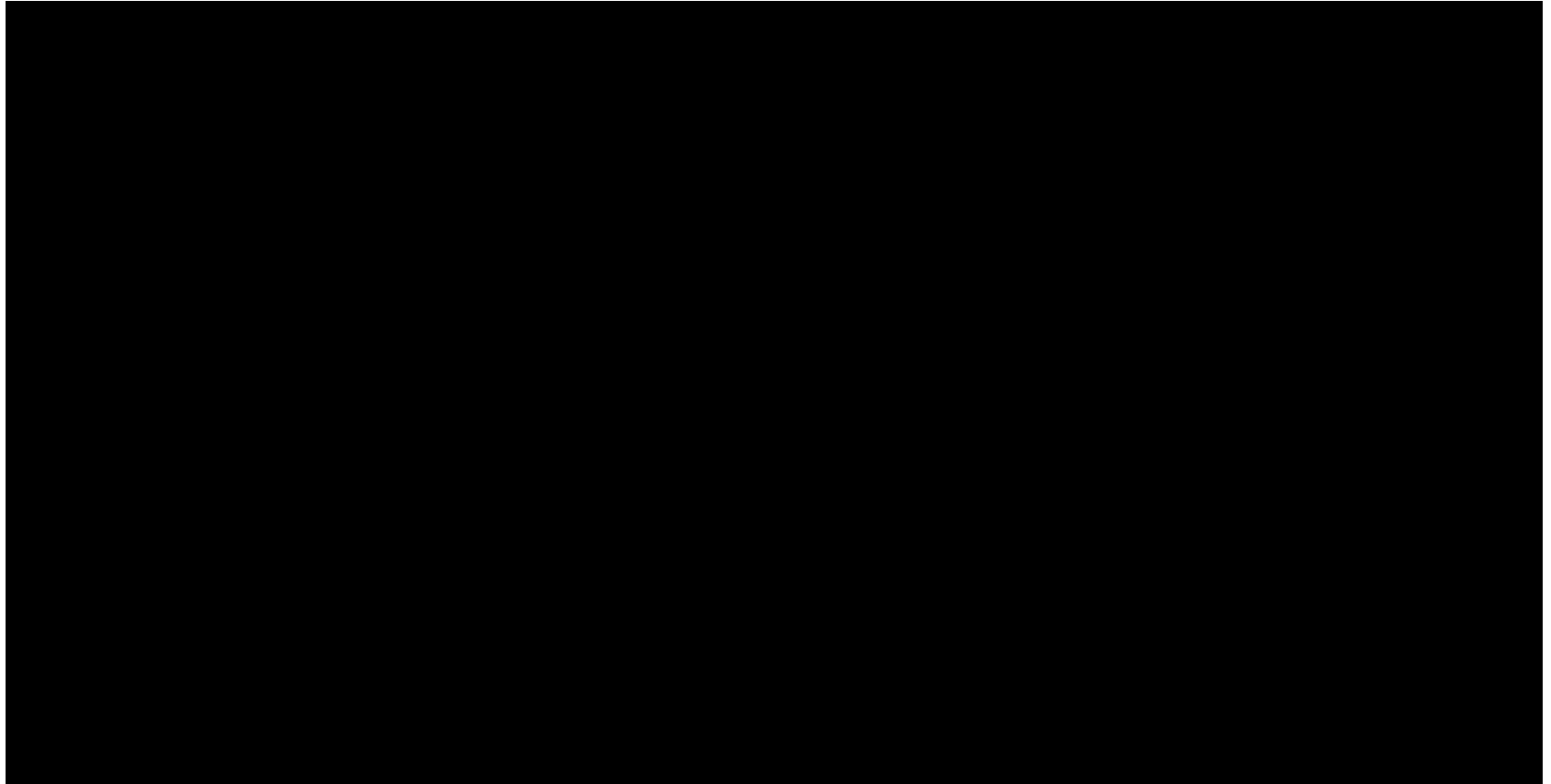
Proposed Method – Work Flow



Describe the process of the project :

The Fire Training and Evaluation Logic is a sophisticated system for comprehensive firefighter training. It includes various modules and interfaces for stakeholders. The public interface educates and raises awareness, while specialized devices enhance firefighter communication and situational awareness. Commanders use dedicated interfaces for real-time monitoring and decision-making. Trainees engage with VR headsets for immersive firefighting scenarios. A central Large Screen/Human Interface Module facilitates collaborative decision-making. The Training Evaluation Module assesses performance, considering response time and SOP adherence. The Training Control Module manages the program systematically. The Real-time VR Engine powers the Fire Training Simulator, simulating lifelike fire scenarios. Integrated Fire Response Guidelines/SOPs guide trainees, and Geometric VR Models and FDS Simulations ensure visual accuracy, enhancing the overall training experience for diverse emergency situations.





Conclusion:

To sum up, the creation of a damage control and firefighting module utilizing virtual reality represents a major breakthrough in emergency response and firefighter training approaches. This module offers a safe yet efficient environment for sharpening critical skills and decision-making abilities by offering realistic and immersive simulations of fire occurrences and threats. Users can practice firefighting skills, deal with dangerous situations, and work as a team in a controlled environment thanks to the interactive aspect of the training. The module provides responders with useful feedback mechanisms and performance evaluation, enabling them to continuously enhance their readiness and efficacy in managing real-world situations. Such cutting-edge training programs are essential to guaranteeing the security and preparedness of firefighters around the globe as technology advances.

Future Work:

Anticipating further developments, the artificial intelligence algorithms that simulate more dynamic fire behavior and environmental conditions may be included into the virtual reality-based firefighting and damage control module. Furthermore, tailoring training sessions according to each user's performance and learning preferences might be achieved by integrating machine learning capabilities. Improved multiplayer capabilities might make it possible to conduct cooperative training exercises with firefighters in several places, which would promote cooperation and teamwork. Additionally, including real-time data feeds from sensors and Internet of Things devices could result in training situations that are more adaptive and realistic, reflecting changing fire dynamics and threats. Using haptic feedback technology in virtual reality could improve realism and immersion by giving firefighters tangible experiences while performing their tasks. These developments could completely change the way that firefighters are trained, making responders more capable of managing the intricacies of crises down the road.

Reference examples

1. H. Kang, J. Yang, B. -S. Ko, B. -S. Kim, O. -Y. Song and S. -M. Choi, "Integrated Augmented and Virtual Reality Technologies for Realistic Fire Drill Training," in IEEE Computer Graphics and Applications, doi: 10.1109/MCG.2023.3303028, Jun 2023.
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3. X. Huang and L. Du, "Fire Detection and Recognition Optimization Based on Virtual Reality Video Image," in IEEE Access, vol. 8, pp. 77951-77961, 2020.
4. P. Monteiro, M. Melo, A. Valente, J. Vasconcelos-Raposo and M. Bessa, "Delivering Critical Stimuli for Decision Making in VR Training: Evaluation Study of a Firefighter Training Scenario," in IEEE Transactions on Human-Machine Systems, vol. 51, no. 2, pp. 65-74, April 2021.
5. Morélot, Sarah & Garrigou, Alain & Dedieu, Julie & Bernard, N'Kaoua. (2021). Virtual Reality for Fire Safety Training: Influence of immersion and sense of presence on conceptual and procedural acquisition. Computers & Education.
6. Moohyun Cha, Soonhung Han, Jaikyung Lee, Byungil Choi, A virtual reality based fire training simulator integrated with fire dynamics data, Fire Safety Journal, Volume 50, Pages 12-24, 2012.
7. Majed, Almejmaj., Jeanine, L., M., Skorinko., Brian, J., Meacham. (2017). The effects of cultural differences between the us and saudi arabia on emergency evacuation—Analysis of self reported recognition/reaction times and cognitive state. Case Studies in Fire Safety.
8. M. Puggioni, E. Frontoni, M. Paolanti and R. Pierdicca, "ScoolAR: An Educational Platform to Improve Students' Learning Through Virtual Reality," in IEEE Access, vol. 9, pp. 21059-21070, 2021.
9. J. C. P. Chan, H. Leung, J. K. T. Tang and T. Komura, "A Virtual Reality Dance Training System Using Motion Capture Technology," in IEEE Transactions on Learning Technologies, vol. 4, no. 2, pp. 187-195, April-June 2011
10. M. S. Anwar, J. Wang, W. Khan, A. Ullah, S. Ahmad and Z. Fei, "Subjective QoE of 360-Degree Virtual Reality Videos and Machine Learning Predictions," in IEEE Access, vol. 8, pp. 148084-148099, 2020.

Outcomes- Participation in Project Contest/Paper Publications

Sl.No.	Roll No.	Name of the Student (s)	Event Title with organizer Details and Date	Prized Won/Participated
1	2202201	RAHUL.P	FIESTAA 2024-KPR institute of engineering and technology (28/3/24)	Participated
2	2202238	SUDHARSHAN.V	FIESTAA 2024-KPR institute of engineering and technology (28/3/24)	Participated
3	2202249	VAGULABARAN.K.M	FIESTAA 2024-KPR institute of engineering and technology (28/3/24)	Participated

THANK YOU